

Lab03

1.

```
library(readr)
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
airports = read_csv("airports.csv",
                    col_select = c(IATA_CODE, CITY, STATE, LATITUDE, LONGITUDE))
```

Rows: 322 Columns: 5

— Column specification —

Delimiter: ","

chr (3): IATA_CODE, CITY, STATE

dbl (2): LATITUDE, LONGITUDE

❗ Use `spec()` to retrieve the full column specification for this data.

❗ Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
airlines = read_csv("airlines.csv")
```

Rows: 14 Columns: 2

— Column specification —

Delimiter: ","

chr (2): IATA_CODE, AIRLINE

❗ Use `spec()` to retrieve the full column specification for this data.

❗ Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
resultado_final = data.frame(
  DESTINATION_AIRPORT = character(),
  soma_atraso = numeric(),
  n_voos = numeric()
)

minha_funcao = function(pedaco, pos) {

  pedaco = pedaco[substr(pedaco$DESTINATION_AIRPORT, 1, 1) != "1", ]
```

```

pedaco = na.omit(pedaco)

resumo_pedaco = pedaco %>%
  group_by(DESTINATION_AIRPORT) %>%
  summarise(soma_atraso = sum(ARRIVAL_DELAY),
            n_voos = n())

resultado_final <- rbind(resultado_final, resumo_pedaco)
}

read_csv_chunked(
  "flights.csv",
  col_types = cols_only(DESTINATION_AIRPORT = col_character(),
                        ARRIVAL_DELAY = col_double()),
  chunk_size = 1000000,
  callback = SideEffectChunkCallback$new(minha_funcao)
)

```

NULL

```

flights = resultado_final %>%
  group_by(DESTINATION_AIRPORT) %>%
  summarise(soma_atraso = sum(soma_atraso),
            n_voos = sum(n_voos)) %>%
  mutate(atraso_medio = soma_atraso / n_voos)

```

2.

a-) Chave em flights: DESTINATION_AIRPORT

b-) Chave em airports: IATA_CODE

c-) Não há aeroportos em flights que não estejam em airports

```

flights %>%
  anti_join(airports, by = c("DESTINATION_AIRPORT" = "IATA_CODE"))

```

```

# A tibble: 0 × 4
#   i 4 variables: DESTINATION_AIRPORT <chr>, soma_atraso <dbl>, n_voos <int>,
#   atraso_medio <dbl>

```

d-) e e-)

```

flights = flights %>%
  left_join(airports %>% select(IATA_CODE, CITY, STATE),
            by = c("DESTINATION_AIRPORT" = "IATA_CODE"))

```

3.

```
airports %>%
  group_by(STATE) %>%
  summarise(n_aeroportos = n()) %>%
  arrange(desc(n_aeroportos))
```

```
# A tibble: 54 × 2
  STATE n_aeroportos
  <chr>   <int>
1 TX         24
2 CA         22
3 AK         19
4 FL         17
5 MI         15
6 NY         14
7 CO         10
8 MN          8
9 MT          8
10 NC         8
# i 44 more rows
```

4.

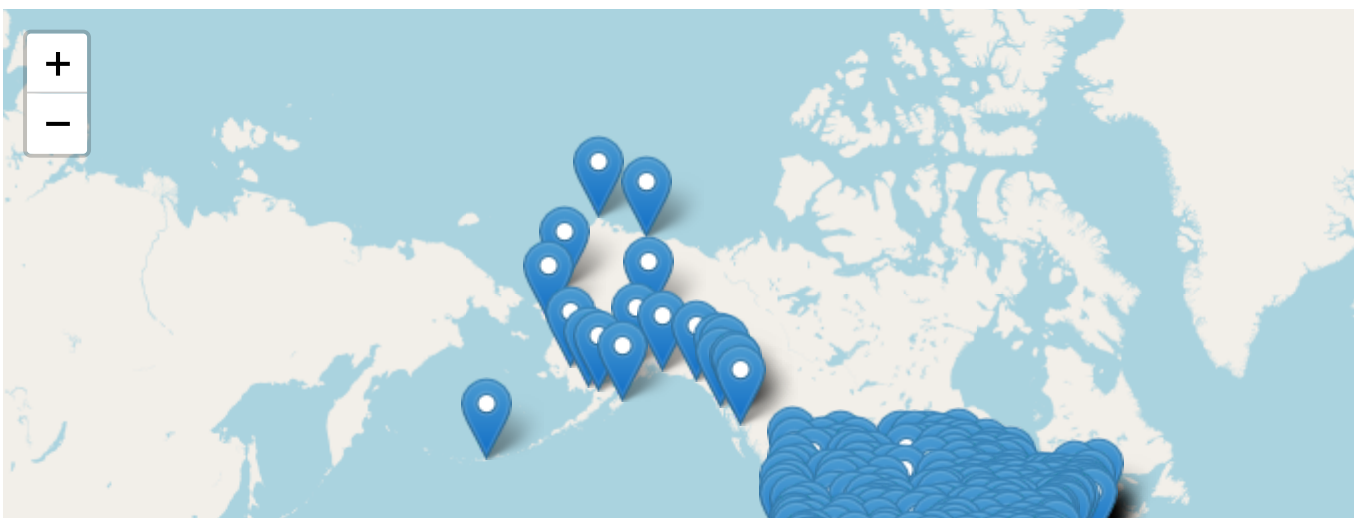
```
library(leaflet)

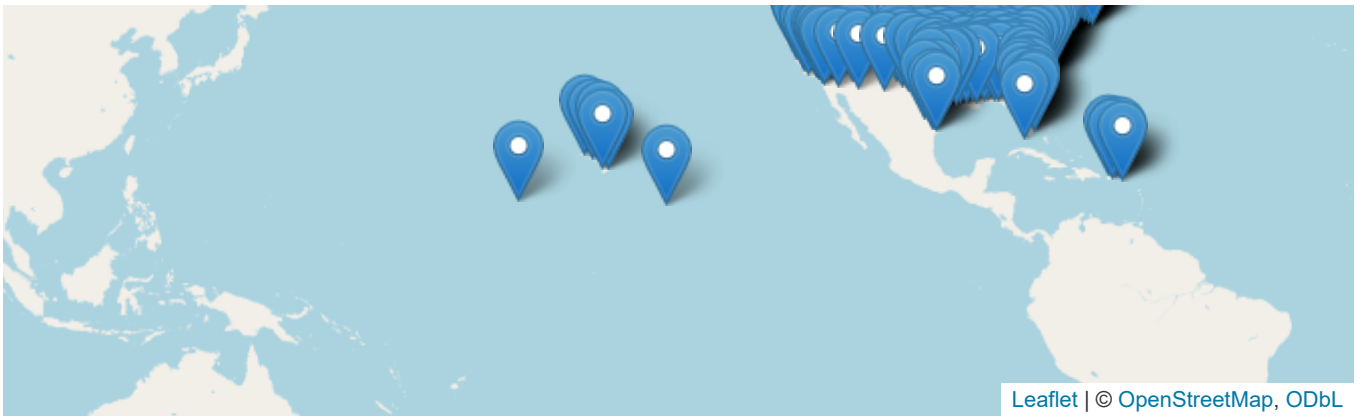
mapa_dados = flights %>%
  left_join(airports %>% select(IATA_CODE, LATITUDE, LONGITUDE),
    by = c("DESTINATION_AIRPORT" = "IATA_CODE"))

this = leaflet(mapa_dados) %>%
  addTiles() %>%
  addMarkers(lng = ~LONGITUDE, lat = ~LATITUDE,
    popup = ~paste(DESTINATION_AIRPORT, "- Atraso médio:", round(atraso_medio, 1)))
```

Warning in validateCoords(lng, lat, funcName): Data contains 3 rows with either missing or invalid lat/lon values and will be ignored

```
this
```





```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-08-31 18:31:52 -03"
```